

$$y = f(x)$$

Def. $f(x) = \sin x$

$$f'(x) = \cos x.$$

P.f. $y' = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$

$$= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$$

$$= \lim_{h \rightarrow 0} \left[\cos x - \frac{\sin h}{h} - \sin x \cdot \frac{1 - \cos h}{h} \right]$$

$\because \sin^2 + \cos^2 = 1$

But. $\lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = \lim_{h \rightarrow 0} \frac{1 - \cos^2 h}{h(1 + \cos h)} = \lim_{h \rightarrow 0} \frac{\sin^2 h}{h(1 + \cos h)} = 0.$

$$\therefore (\sin x)' = \cos x.$$

Def. $y = \cos x$

$$y' = -\sin x$$

P.f. $\cos x \Leftrightarrow \sin\left(\frac{\pi}{2} + x\right)$ $u = \frac{\pi}{2} + x.$

$$y' = \left[\cos\left(\frac{\pi}{2} + x\right) - \left(\frac{\pi}{2} + x\right)' \right] = -\sin x$$

$$\therefore (\cos x)' = -\sin x.$$